

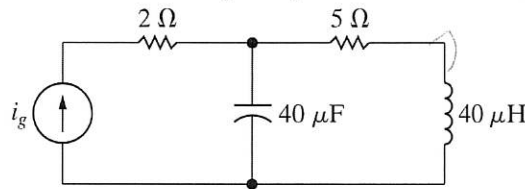
ECE 3364 – Circuits and Systems Quiz #1 Spring 2026

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Problem (100 points). show all work for full credit. Express your answers with appropriate units (2' for each).



Ideal current source $i_g = 30 \cos(25000t)$ mA

$$I = \frac{30}{\sqrt{2}} =$$

$$5 + (j) = 5 + j$$

$$5 + j \parallel -j = 0.2 - j$$

Questions:

(a) (10') What is the rms value I_{rms} of the current source?

$I_{rms} =$ 21.213 ✓, unit mA ✓

(b) (10') What is the angular frequency of current source?

$\omega =$ 25000 ✓, unit rad/sec ✓

(c) (10') What is the impedance of the capacitor?

$Z_c =$ $-j$ ✓, unit Ω ✓

(d) (10') What is the impedance of the inductor?

$Z_L =$ j ✓, unit Ω ✓

(e) (20') What is the equivalent impedance Z_{eq} seeing from the source?

$Z_{eq} =$ $2.2 - j$ ✓, unit Ω ✓

(f) (20'+10'+10') What is the complex power S / average power P / reactive power Q delivered by the ideal current source?

$S =$ $989.981 - 449.991j$ ✓, unit μVA ✓

$P =$ 990 ✓, unit μW ✓

$Q =$ -450 ✓, unit μVAR ✓

$$S = |I_{rms}|^2 Z$$

$$P = V_{rms} \cdot I_{rms} \cdot \cos(\theta_v - \theta_i)$$

$$Q = V_{rms} \cdot I_{rms} \sin(\theta_v - \theta_i)$$

$$Z = \frac{V}{I} = 1 \cdot Z = V = (21.213)(2.2 - j)$$